

Discrimination and Memory

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Motivation

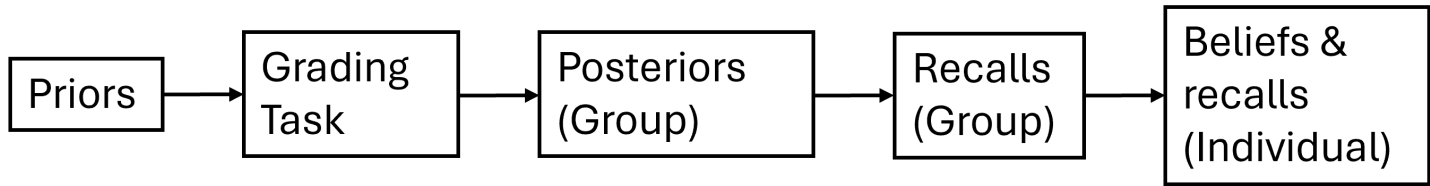
Discrimination is well-documented, and persists over time

Motivation for discrimination: taste-based or statistical → inaccurate beliefs

Why do inaccurate beliefs persist? → **memory on perceived gender difference**

- Belief update is asymmetric (Sarsons, 2022)
- Experience is endogenous (Lepage, 2024)
- Representative heuristics (stereotypes) (Bordalo et al., 2016)
- Misspecified model (Bohren et al., 2019)

Experimental Design



Grading Task: (constructed from Bordalo et al. (2019))

- 5 car-related multiple-choice questions (“male-typed”)
- 20 US university students, 10 males and 10 females

Treatments: (2 × 2 between-subject design)

- **Show/Hide:** scores in the student list
- **Sample:** females outperform males, or vice versa

Students: (actual gender with frictional names)

- 10 males, 10 females with at least 0.5 score difference in mean
- Exactly 1 male and 1 female having scores of 0 and 5, respectively (“outliers”)
- Fix 4 outliers at 4th, 8th, 12th, 16th (randomize)

Elicitation: 1) Average group performance, 2) Gender difference

Grading Task – Example

Grading Task (1/20)

Name: OliviaGender: Female

Grading

The term ‘cabriolet’ attached to a model name would indicate that _____.

☐ The car is an upscale luxury variant.

☒ The car is a convertible; having a folding roof.

☐ The car is a sport-oriented performance variant.

☐ The car has only two doors.

☐ The car is European.

Which of the following boutique automakers originated in Sweden?

☐ Spyker

☐ Koenigsegg

☐ Pagani

☐ Wiesmann

☒ Bugati

What car company owns the Malibu?

☐ Chevrolet

The car is a convertible; having a folding roof.

☐

Koenigsegg

☐

The Hide treatment

Gender	Name	Graded	Name	Graded
Male	Andrew	✓	Joseph	✓
Male	Christopher	✓	Joshua	✓
Male	Daniel	✓	Matthew	✓
Male	Ethan	✓	Michael	✓
Male	Jacob	✓	William	✓

Gender	Name	Graded	Name	Graded
Female	Ashley	✓	Isabella	✓
Female	Elizabeth	✓	Jenny	✓
Female	Emily	✓	Madison	✓
Female	Emma	✓	Olivia	✓
Female	Hannah	✓	Samantha	✓

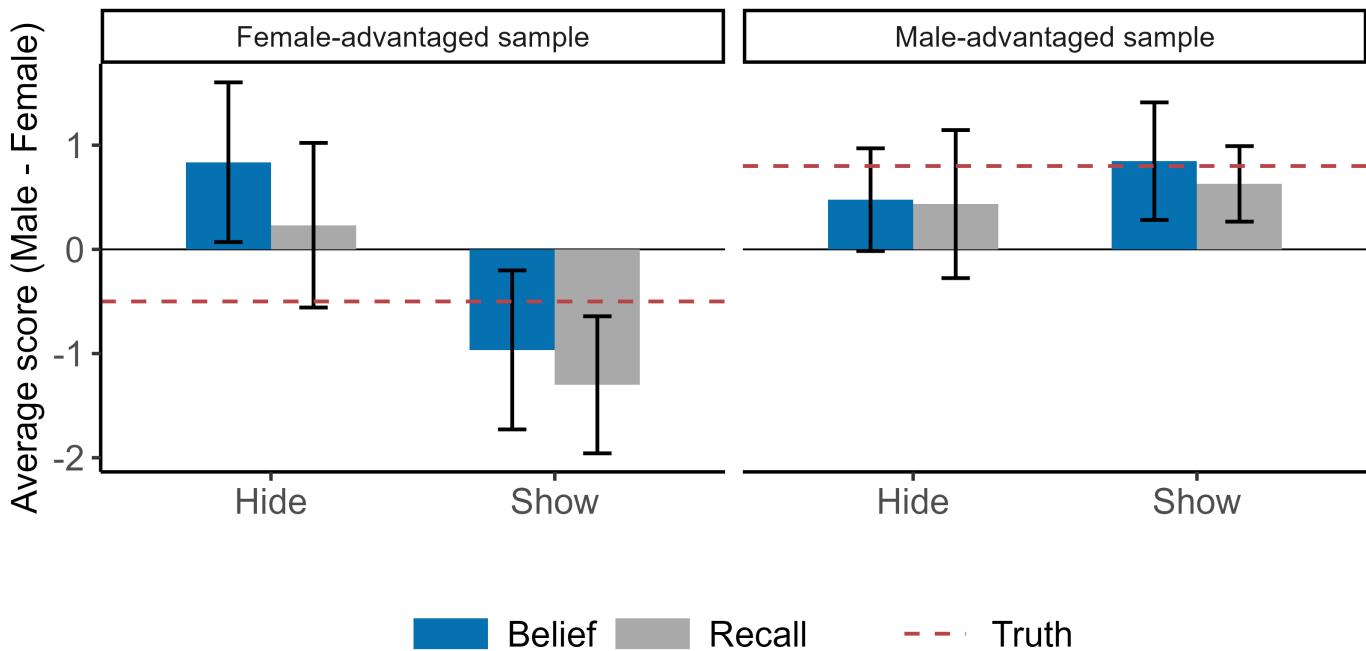
The Show treatment

Gender	Name	Graded	Name	Graded
Male	Andrew	1	Joseph	2
Male	Christopher	4	Joshua	1
Male	Daniel	0	Matthew	3
Male	Ethan	2	Michael	4
Male	Jacob	5	William	4

Gender	Name	Graded	Name	Graded
Female	Ashley	0	Isabella	1
Female	Elizabeth	2	Jenny	5
Female	Emily	1	Madison	2
Female	Emma	2	Olivia	1
Female	Hannah	2	Samantha	2

Results from pilot

Result 1: Biased imperfect recall for gender difference

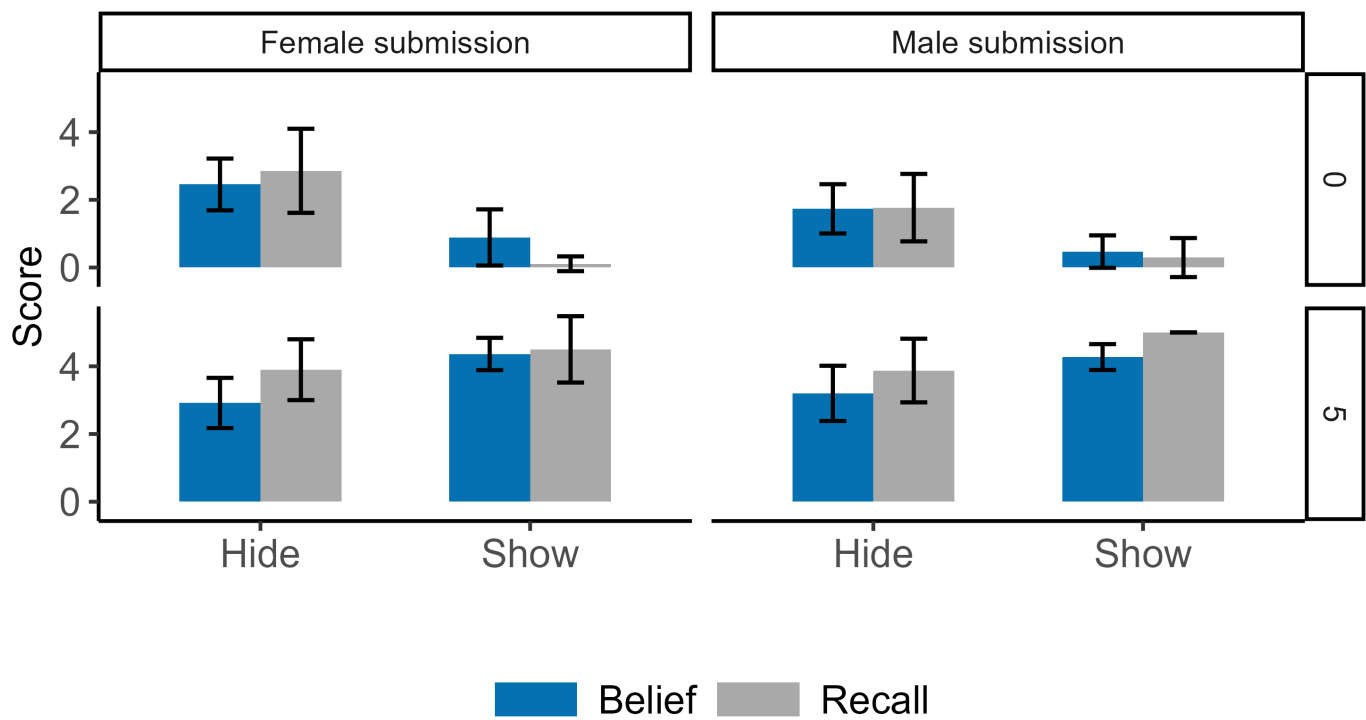


Result 2: Symmetric average score belief & recall

	Dependent variable:							
	Difference (Recall)		Difference (Belief)		Average (Recall)		Average (Belief)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
F-Sample × Hide	1.73** (0.65)	1.73** (0.66)	2.17*** (0.65)	2.20*** (0.64)	0.02 (0.27)	−0.05 (0.25)	0.48 (0.35)	0.42 (0.34)
Hide	−0.19 (0.43)	−0.19 (0.44)	−0.37 (0.43)	−0.32 (0.42)	0.21 (0.18)	0.18 (0.16)	−0.20 (0.23)	−0.22 (0.23)
F-Sample	−1.93*** (0.46)	−1.93*** (0.47)	−1.81*** (0.46)	−1.83*** (0.45)	0.34* (0.19)	0.32* (0.18)	0.26 (0.25)	0.24 (0.24)
Constant	0.63** (0.30)	0.61* (0.35)	0.85*** (0.30)	0.55 (0.33)	2.33*** (0.12)	1.74*** (0.23)	2.31*** (0.16)	1.82*** (0.32)
Prior	No	Yes	No	Yes	No	Yes	No	Yes
Observations	48	48	48	48	48	48	48	48

Note: *p<0.1; **p<0.05; ***p<0.01

Result 3: Symmetric individual beliefs & recalls



Conclusion

- Discrimination persists only when people need to rely on memory
- This is driven by biased imperfect recall in group statistics
- No gender difference at the individual level (recall, belief)
- Discrimination happens for repeated cross-sectional, but not in individual panel

